

Simulating Memory

Teaching Gullah-Geechee Heritage through Virtual Performances by Community Leaders at a Reconstruction-Era Historical Site

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Abstract

NCSU Media Arts, Design and Technology (MADTech) faculty collaborated with Historic Mitchelville Freedom Park on a sponsored studio course that taught students digital heritage research by developing digital double scenes of Gullah-Geechee community leaders in virtual park landmarks. Founded in 1862, Mitchelville was the first self-governing town of enslaved Africans emancipated by the Union army during the American Civil War and currently thrives as a community-run Gullah-Geechee heritage site on Hilton Head Island, South Carolina. In the spring 2025 semester, MADTech faculty taught students how to develop animated digital doubles using photogrammetry and performance capture in our College of Design Virtual Production Lab and then brought students to Hilton Head Island over spring break to gather community and location data at the park. Faculty arranged community tours of local monuments and we scanned a new “Ghosted Structures” interpretive exhibit of where archeologists uncovered Mitchelville homes. Students assisted faculty in 360 photography of the park and spatially recording Gullah elders and park leaders describing what the park meant to them. MADTech students considered ethical questions related to digital double representations of living community leaders and strived to amplify community voices through MetaHuman monologues using Unreal Engine, UEFN and gen-AI tools. Students also gained valuable virtual production skills in exploring digital heritage applications in a sponsored research project and earned college global experience credits by learning about Gullah-Geechee culture from the descendants of the Mitchelville community.

CCS Concepts

• Applied computing; • Education; Arts and humanities; • Media arts;

Keywords

Digital Heritage, Digital Doubles, Virtual Production, Generative AI

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1 Introduction

Media Arts, Design and Technology (MADTech) faculty co-taught a sponsored studio in spring semester 2025 where upper-level MADTech and Design Studies students learned digital double applications for cultural heritage research in our College of Design Virtual Production Lab. Students then visited a Gullah-Geechee heritage site over spring break to gather spatial field data for developing virtual scenes with digital doubles of community leaders and elders. This special-topics course furthered digital double curriculum presented in SIGGRAPH 2024 Educator’s Forum where we showed MetaHuman monologues created by MADTech students in Unreal Engine [Maraffi, 2024]. Our spring 2025 MADTech studio course was sponsored by Historic Mitchelville Freedom Park (HMFP) on Hilton Head Island South Carolina, an active Gullah-Geechee community monument “where freedom began” as the first self-governing town of emancipated Africans during the American Civil War. Founded in 1862 as part of the Union’s Port Royal Experiment that was a test-model for Reconstruction after the war [Barnwell et al., 2020], Mitchelville was inhabited by emancipated people from coastal plantations in the Carolina and Georgia Sea Islands known today as the Gullah-Geechee corridor.

During the first half of the course, students learned about the history of Mitchelville and its African Diaspora culture through readings in our Gullah Days textbook [Barnwell et al., 2020]. They also learned how to apply emerging technologies like photogrammetry, performance capture and generative AI to creating digital doubles of themselves. To design 3D likenesses of their heads, students shot cell phone video of each other and processed footage to a photogrammetry mesh using Epic Games’ Reality Scan. They used their scans to develop a rigged digital double in Unreal Engine 5.4 using MetaHuman Creator and Animator, and animated their MetaHumans using performance capture and gen-AI tools. Prior to creating MetaHumans, students experimented with various generative AI workflows to improve the visual quality of digital doubles and better understand how they can be integrated into immersive production environments. They used CrAIyon, Runway, RADiCAL, and Blockade Labs AI tools to visualize scenes, motion capture with a webcam, and create equirectangular backgrounds. In preparation for visiting Mitchelville over spring break, students used 360 cameras for capturing real-world locations around our



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Table 1: Metadata in tabular format.

Summary	Students receive global experience credits by learning about Gullah-Geechee culture while conducting applied design field research in emerging technologies.
Learning Outcomes	Students learn to apply a virtual production pipeline to design digital doubles in immersive scenes related to Mitchelville historical site and Gullah-Geechee culture.
Classification(s)	Applied computing, Arts and Humanities, Media Arts.
Audience	Undergraduate students in media arts, digital heritage and design studies.
Dependencies	Students were required to participate in the spring break field trip to the site to receive global experience credits.
Prerequisites	No prerequisites were required.
Strengths	Students gained hands-on experience with applying a variety of emerging technologies to develop ethical digital double scenes while learning Gullah-Geechee heritage.
Weaknesses	Requires substantial organization and planning to visit the site and conduct field research in the community.
Variants	Early exercises allowed students to learn the process with creative variations.
Assessment	Students were assessed through weekly individual assignments and a final group project and research poster.

Raleigh campus as high dynamic range (HDR) images for use as backdrops in Unreal Engine.

In the second half of the semester, students and faculty travelled to Hilton Head Island over spring break to conduct field research and gather data for developing a virtual tour scene of “ghosted structures” recently installed in the park. The week-long field trip was required for students to receive college global experience credits by learning about Gullah-Geechee culture through heritage tours by community members on Hilton Head Island and at the nearby *Reconstruction Era National Historical Park* in Beaufort.

Field research included 360 photography of park sites and spatial recordings of HMFP Director Ahmad Ward introducing the Ghosted Structures interpretive tour, Gullah Days co-author and islander descendent Carolyn Grant talking about Mitchelville and the Port Royal Experiment, and award-winning Gullah performer Dr. Marlena Smalls singing *Amazing Grace*. Students helped to gather performance capture data using a Noitom inertial motion capture suit and a Rokoko head rig with MetaHuman Animator face capture app. Immersive design frameworks we used for scene development in Unreal Engine included Curtis Hickman’s hyper-reality principles used to develop the VOID VR experiences, which students learned from several guest lectures from Hickman and through his textbook [Hickman, 2023].

2 Course materials

Please contact Topher Maraffi for the 2025 special topics course syllabus and MAGIC Tour of Mitchelville project website info.

3 Results and Discussion

Focusing on performing digital doubles for a virtual heritage application allowed us to discuss ethical issues of representation, from a broader contemporary performance perspective of recent Screen Actors Guild strikes [Sag, 2023] and more specifically from representing Gullah-Geechee community leaders with hyper-real MetaHumans and gen-AI. Rather than try to academically simulate famous figures like Harriet Tubman who worked on Hilton

Head Island as a mediator between Union soldiers and the Gullah people that founded Mitchelville, we are simulating performers in the park who impersonate Tubman at heritage festivals through the community storytelling tradition. To facilitate preserving the community’s collective memory of Mitchelville, we have shifted focus to representing living leaders and elders in the community to help them tell their story to a wider audience through emerging technologies. This allows us to stay close to the community source material while minimizing our own outside interpretations. Elder performers like Marlena Smalls can better interpret the importance of Gullah hymns and praise houses for the original residents of Mitchelville. Since our trip to HMFP, students and faculty have worked on postproduction processing of the Ghosted Structures scene to port Mitchelville’s community story into a variety of immersive platforms, including a live interactive experience in an LED wall for use in the park, and a UEFN adaptation to develop a Fort-nite Mitchelville Island that will bring the community memories to a new audience.

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References

Hickman, Curtis. 2023. *Hyper-Reality: The Art of Designing Impossible Experiences*. Independently published on Amazon.
Barnwell, Thomas et al. 2020. *Gullah Days: Hilton Head Islanders Before the Bridge 1861-1956*. Blair; Illustrated Edition.
Maraffi, Topher. 2024. *Metahuman Theatre: Teaching Photogrammetry and Mocap as a Performing Arts Process*. SIGGRAPH 2024 Educator’s Forum.
Sag, Aftra. 2023. *SAG-AFTRA: Statement on the Use of Artificial Intelligence and Digital Doubles in Media and Entertainment*. News Update March 17. .